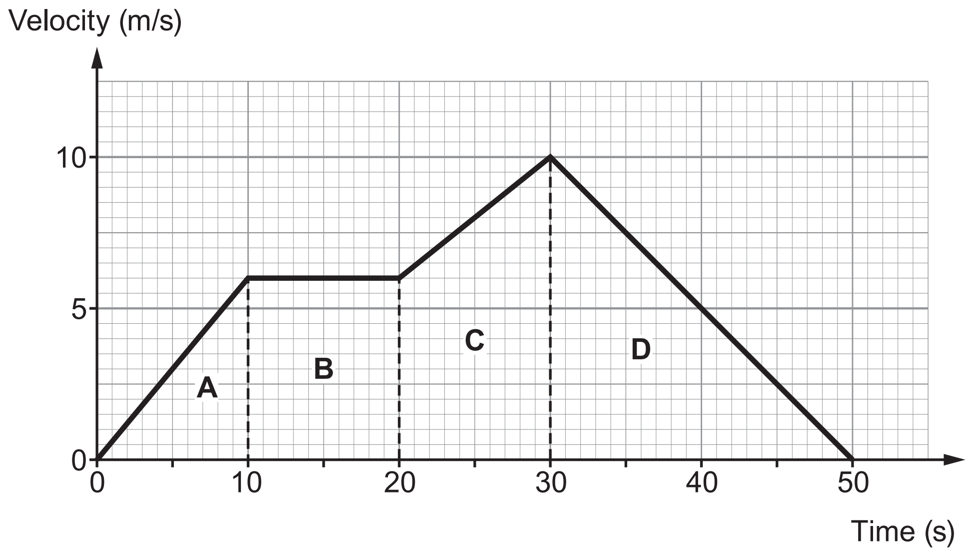
Examiner only

2. The velocity-time graph is for part of a bus journey.



Use the information in the graph to answer the following questions.

(a) **Complete the table** by placing **one** tick (✓) in each row to describe the motion in each region of the graph. Region A has been completed as an example. [3]

Region of graph	Not moving	Constant velocity	Accelerating	Decelerating
A			✓	
B				
C				
D				

(b) **Complete the following sentences** using numbers from the box. [3]

2	4	6	8	10	20	50

(i) The maximum velocity of the bus is .... m/s.

(ii) The change in velocity of the bus in region C is .... m/s.

(iii) The bus accelerates for a **total time** of .... s.

(c) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration in **region A**. [3]

acceleration = ... m/s^2

(d) The bus travelled 270 m in the 50 s shown.

Use the equation:

$$\text{mean speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the mean speed of the bus. [2]

mean speed = ... m/s

11